Service Association and Subcarrier Allocation in Vehicular Slicing Networks With Dynamic Multi-Service

Yijian Hou, Kaisa Zhang, Gang Chuai, Weidong Gao, Xiangyu Chen, and Siqi Liu

***Abstract*— The vehicular slicing network with dynamic multi-service (VSN-DMS) enables vehicles to support multiple services with dynamically changing types. In this architecture, vehicles can access multiple base stations (BSs) and leverage different network slices (NSs) to meet the requirements of various services. However, frequent service variations and vehicle mobility introduce high handover (HO) costs and inefficient decision making. This letter proposes a novel two-timescale optimization framework to jointly optimize vehicle service (VS) association and subcarrier (SC) allocation. Specifically, in the long-timescale, we propose a federated deep reinforcement learning (FDRL) algorithm to optimize VS association. Unlike conventional reinforcement learning approaches that require centralized data collection, FDRL enables distributed model training across vehicles while aggregating parameters in the cloud. This mitigates the issue of insufficient training data caused by frequent VS changes, while also ensuring vehicle data privacy. In the short-timescale, we propose a versatile alternating coalitional game and swap-matching game algorithm (ACGSM) to optimize SC allocation. By iteratively executing coalitional formation and swap-matching mechanisms, ACGSM efficiently improves the quality of the solution while adapting to dynamic network conditions. Simulation results demonstrate the convergence of FDRL and show that the proposal outperforms benchmarks in improving long-term QoS and reducing the number of HOs.**

***Index Terms*— Vehicular slicing network, two-timescales, service association, subcarrier allocation, federated deep reinforcement learning, game theory.**

I. Introduction

IN VEHICULAR networks, vehicles function as multi-service terminals that support various services simultaneously and dynamically adjust service types based on demand. To meet the quality of service (QoS) requirements of different vehicle services (VSs), vehicles can connect to multiple BSs and receive customized services through network slicing technology, which virtualizes network resources into multiple logical network slices (NSs) to serve different service categories with customized QoS guarantees. We refer to this architecture as "vehicular slicing network with dynamic multi-service (VSN-DMS)". In VSN-DMS, the association of VSs with base stations (BSs) and the issue of subcarrier (SC) allocation needs to be addressed. Moreover, performing VS association on a short timescale may lead to frequent handovers (HO) and excessive signaling overhead. Considering the dynamic nature of wireless networks, resource allocation strategies must be both rapid and flexible. Therefore, the above issues need to be addressed in different time scales. Next, we will detail the challenges of addressing these two issues and our solutions.

First, for the VS association issue, vehicles with multi-connectivity capability can connect to different NSs to access multi-services. For instance, the work in [1] addressed the association of two different services in vehicular slicing network (VSN), where the association scheme is determined based on optimal channel state information (CSI). However, relying solely on CSI to determine the access point is "one-sided". Therefore, the work in [2] proposed a multi-attribute decision-making approach that considered the delay, packet loss rate, and energy consumption of candidate BSs. Nevertheless, this work assumed prior knowledge of the QoS provided by BSs, whereas in reality, QoS is influenced by multiple factors such as load and HO frequency. Besides, Users should prioritize long-term QoS over instantaneous performance. Considering this, the work in [3] used deep reinforcement learning (DRL) to optimize the long-term QoS of devices by considering factors such as load and HO cost. But it neglected the impact of historical association schemes and vehicle mobility on subsequent association decisions. Accordingly, we propose a DRL algorithm that incorporates vehicles' mobility state, HO cost, load, and available resources as states to derive the association scheme for VSs. Besides, in dynamic multi-service (DMS) scenarios, the occurrence and duration of VSs are uncertain. When a new service request arises, it needs to promptly select a BS based on the current network conditions. If the DRL model starts training at this moment, early decisions may perform poorly due to insufficient training data. To address this, we introduce the federated learning, which retains training data on local devices and learns a global model by aggregating model parameters [4]. Using federated DRL (FDRL), we can improve the convergence of the DRL model while preserving the data privacy of vehicles.

Second, for the SC allocation issue, this letter focuses on the real-time intra-slice SC allocation problem, assuming that the resources of NSs have already been determined. Although some studies have addressed this issue, there are still some shortcomings. For example, the work in [5] transformed the SC allocation problem into a convex optimization problem and adopted the alternating direction method of multipliers

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Yijian Hou, Kaisa Zhang, Gang Chuai, and Weidong Gao are with the School of Information and Communication Engineering, Beijing University of Posts and Telecommunications, Beijing 100876, China (e-mail: houyijian@bupt.edu.cn; kaisa@bupt.edu.cn; chuai@bupt.edu.cn; gaoweidong@bupt.edu.cn).

Xiangyu Chen is with the School of Electronic Engineering, Beijing University of Posts and Telecommunications, Beijing 100876, China (e-mail: xychen324@bupt.edu.cn).

Siqi Liu is with the School of Electrical and Electronic Engineering, North China Electric Power University, Baoding 071003, China (e-mail: liusiqi@ncepu.edu.cn).

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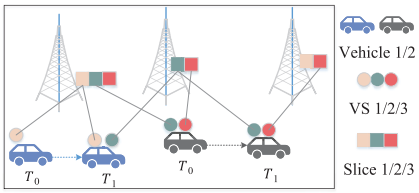


Fig. 1. Dynamic Multi-Service Vehicle Slicing Network Diagram.

to solve it. However, optimization algorithms like this have strict restrictions on the form of the objective function and do not support the design of personalized utility functions for different VSs. To avoid such limitations, some studies have adopted DRL to solve the SC allocation problem. However, DRL generally requires that the state space and action space remain unchanged, which is not applicable in dynamically changing communication networks. In light of these challenges, we introduce the idea of game theory and propose an alternating coalitional game and swap-matching game algorithm (ACGSM) which imposes no restrictions on the form of the objective function or the variability of the state and action spaces.

Briefly, the main contributions include: 1) We are the first to propose a joint VS association and SC allocation scheme (VSASA) based on two-timescales in VSN-DMS. 2) For VS association in long-timescale, we propose an FDRL algorithm that takes into account the vehicles' mobility, HO cost, load and other factors. The algorithm is proved to improve the long-term QoS of VSs and reducing the number of HOs. 3) For SC allocation in short-timescale, we propose ACGSM, which is versatile and is proved to yield near-optimal solutions.

II. System Model

We consider the orthogonal frequency division multiple access downlink vehicular network system as shown in Fig. 1, in which there are different NSs on the BS to provide services for the corresponding VSs. Let $\mathcal{M} = \{1, \ldots, m, \ldots, M\}$ and $\mathcal{V} = \{1, \ldots, v, \ldots, V\}$ represent the sets of BSs and vehicles, respectively. $\mathcal{N} = \{1, \ldots, n, \ldots, N\}$ denotes the set of NSs, which has a one-to-one correspondence with the service types. Each vehicle supports one or more services and has multi-connectivity capability. Different services in the same vehicle can be connected to different BSs through wireless links to receive services simultaneously [1]. We use vector $\boldsymbol{y}$ to represent the association indicator, such that $y_{v,n}^m = 1$ indicates that the service n in vehicle v ($\text{VS}_{v,n}$) is associated with BS m. The number and types of services for each vehicle may vary over time. For instance, in Fig. 1, vehicle v_1 initially has only service s_1 active at time T_0 and then activates an additional service s_2 at time T_1.

When a VS is associated with an NS in a certain BS, the NS needs to allocate SCs to this VS. We use vector $\boldsymbol{x}$ to represent the SC allocation indicator, such that $x_{v,n}^{m,c} = 1$ indicates that NS n in BS m allocates SC c to $\text{VS}_{v,n}$. Therefore, the data rate of $\text{VS}_{v,n}$ can be expressed as

$$r_{v,n} = \sum_{m=1}^{M} y_{v,n}^m \sum_{c=1}^{C} x_{v,n}^{m,c} W_0 \log_2 \left(1 + \frac{P_m g_{v,n}^{m,c}}{I_{v,n}^{m,c} + \sigma^2}\right), \quad (1)$$

where W_0 is the bandwidth of each SC, and P_m is the transmitted power. $g_{v,n}^{m,c} = P_L^{m,v} \xi_{v,n}^{m,c} h_{v,n}^{m,c^2}$ represents the channel gain, which includes path loss $P_L^{m,v}$, shadow fading $\xi_{v,n}^{m,c}$ that follows log-normal distribution, and fast fading coefficient $h_{v,n}^{m,c}$ that follows first-order Gaussian Markov process [6]. $I_{v,n}^{m,c} = \sum_{m' \in \mathcal{M} \setminus m} \sum_{c \in \mathcal{C}} \sum_{v' \in \mathcal{V}} \sum_{n' \in \mathcal{N}} x_{v',n'}^{m',c} P_{m'} g_{v',n'}^{m',c}$ denotes the received interference power. In this letter, we use data rate and transmission delay as QoS metrics, where the delay is calculated as $d_{v,n} = b_n / r_{v,n}$ and b_n is the packet size. We use $s_{v,n}^r = 1/\left(1 + \exp\left(\frac{-\alpha(r_{v,n} - r_n^{min})}{r_n^{max} - r_n^{min}} + \beta\right)\right)$ and $s_{v,n}^d = 1/\left(1 + \exp\left(\frac{-\alpha(d_n^{max} - d_{v,n})}{d_n^{max} - d_n^{min}} + \beta\right)\right)$ to represent the satisfaction of $\text{VS}_{v,n}$ for data rate and delay, respectively [7]. α and β are used to control the sensitivity and benchmark value of the utility function, respectively. Thus, the satisfaction of $\text{VS}_{v,n}$ is defined as $S_{v,n} = w_n^r s_{v,n}^r + (1 - w_n^r) s_{v,n}^d$, where w_n^r represents the weight of data rate. It is defined as a tunable parameter ranging from [0,1], where a higher value indicates that the service places greater importance on data rate. The utility of $\text{VS}_{v,n}$ is denoted as $U_{v,n} = S_{v,n} - \gamma G_{v,n}$, where $G_{v,n}$ is a function related to the number of allocated SCs and the satisfaction, aiming to avoid excessive SCs allocation. It is defined as $G_{v,n} = \sum_{m=1}^{M} \sum_{c=1}^{C} x_{v,n}^{m,c} / \left(1 + \exp\left(\frac{-\alpha(S_{v,n} - S_{min})}{1 - S_{min}} + \beta\right)\right)$, where S_{min} represents the minimum acceptable satisfaction.

In this letter, we introduce a two-timescales model to address the VS association and SC allocation problem. In the long-timescale, the associated BS needs to be determined for each VS. The conditions for triggering the decision action include: 1) $\Delta t_{\max}$ short timeslots have passed since the last association decision; 2) the reference signal receiving power (RSRP) is below a threshold; 3) the VS is activated in this time slot. We use T to denote the the number of long timeslots, and Δt_T to represent the number of short timeslots between the T-th and $(T+1)$-th long timeslots. t_T is used to denote the number of short timeslots that have passed since the T-th long timeslot. The objective for each VS in this stage is defined as

$$\mathcal{P}1: \max_{\mathbf{y}_{v,n}} \sum_{T=0}^{\infty} \left(\frac{1}{\Delta t_T} \sum_{t_T=0}^{\Delta t_T} U_{v,n}^{t_T} - HC^T\right) \quad (2a)$$

$$\text{s.t. } y_{v,n}^m \in \{0,1\}, \ \forall m \in \mathcal{M} \quad (2b)$$

$$\sum_{m=1}^{M} y_{v,n}^m = 1, \quad (2c)$$

HC^T is the HO cost, which is set to 0 if no HO occurs and to a fixed value within $(0,1)$ when a HO takes place.

In the short-timescale, each NS needs to allocate SCs to the associated VSs. At each short timeslot t_T, the objective for each NS is expressed as:

$$\mathcal{P}2: \max_{\mathbf{x}_{m,n}} (1 - w_f) \frac{\sum_{v=1}^{V} y_{v,n}^m U_{v,n}^{t_T}}{\sum_{v=1}^{V} y_{v,n}^m} + w_f J(m,n) \quad (3a)$$

$$\text{s.t. } x_{v,n}^{m,c} \in \{0,1\}, \ \forall v \in \mathcal{V}, c \in \mathcal{C} \quad (3b)$$

$$x_{v,n}^{m,c} \le y_{v,n}^m z_{m,n}^c, \ \forall v \in \mathcal{V}, c \in \mathcal{C} \quad (3c)$$

$$\sum_{v=1}^{V} x_{v,n}^{m,c} \le 1, \quad (3d)$$

where $z_{m,n}^c = 1$ denotes that SC c in BS m is assigned to slice n. (3d) limits that each SC is allocated to at most one

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VS. To ensure fairness in the NS, we introduce the fairness index $J(m,n)$, with w_f as the importance of fairness. $J(m,n)$ is calculated using the Jain's fairness index [8].

III. The Proposed Two-Timescales Scheme

To address the challenges outlined in Section I, we propose a two-timescale approach in this section. Next, we will introduce this approach in conjunction with **Algorithm 1**.

A. FDRL-Based VS Association Scheme in Long-Timescale

Considering that $\mathcal{P}1$ aims to maximize the long-term utility of the VS, we introduce the discrete-action DRL algorithm, i.e., double deep Q-Network (DDQN). DDQN is well-suited for optimizing long-term rewards through sequential decision-making. By introducing experience replay and target networks, it effectively handles large state-action spaces, demonstrating superior training performance [3]. Additionally, we should address two key challenges: 1) In DMS scenarios, vehicles may suddenly generate new service demands, for which the experience replay buffer may lack sufficient data for online training; 2) Traditional DRL requires collecting vehicle data for centralized training, which increases the risk of privacy breaches. Vehicle data includes location information, service type, network status, etc. If such data is exploited by malicious attackers, it could degrade communication quality or even pose security threats to users. To address these challenges, we adopt the FDRL algorithm, where vehicles train their models locally and only need to upload model weights. The central server aggregates these parameters without accessing raw data, which protects data privacy while enhancing training efficiency [3].

In detail, each VS selects a BS from the list of candidate BSs ($\mathcal{K}_{v,n}^T$) when it needs to make an association decision. Each VS has a DDQN agent, with its state, action, and reward defined as follows:

- State ($s_{v,n}^T$): the distances between candidate BSs and vehicle v; the cosine values of the angles between vehicle v's moving direction and the lines connecting it to candidate BSs; the numbers of users associated with NS n in the candidate BSs; the numbers of SCs of NS n in the candidate BSs; the speed of vehicle v; the average utility of vehicle v since its last association decision, i.e., $\frac{1}{\Delta t_{T-1}}\sum_{t_{T-1}=0}^{\Delta t_{T-1}} U_{v,n}^{t_{T-1}}$.
- Action ($a_{v,n}^T$): the location of the associated BS in $\mathcal{K}_{v,n}^T$.
- Reward ($R_{v,n}^T$): $\frac{1}{\Delta t_T}\sum_{t_T=0}^{\Delta t_T} U_{v,n}^{t_T} - HC^T$.

The sorting principle in $\mathcal{K}_{v,n}^T$ is: 1) the currently associated BS is always ranked first; 2) the remaining BSs are sorted in order from the nearest to the farthest in terms of distance.

DDQN contains two neural networks, namely the evaluation network and the target network. They have the same structure but different parameters, i.e., $\boldsymbol{\theta}$ and $\hat{\boldsymbol{\theta}}$. In each training round, we store the experience ($s_{v,n}^T$, $a_{v,n}^T$, $R_{v,n}^T$, $s_{v,n}^{T+1}$) in the replay memory buffer and randomly sample a mini-batch from the buffer for training. The evaluation network parameters are updated using the following loss equation:

$$L(\boldsymbol{\theta}_{v,n}) = \mathbb{E}[(R_{v,n}^T + \gamma Q(s_{v,n}^{T+1}, \arg\max_{a_{v,n}^{T+1}} Q(s_{v,n}^{T+1}, a_{v,n}^{T+1}; \boldsymbol{\theta}_{v,n}); \hat{\boldsymbol{\theta}}_{v,n}) - Q(s_{v,n}^T, a_{v,n}^T; \boldsymbol{\theta}_{v,n}))^2], \quad (4)$$

Algorithm 1 The Proposed Two-Timescales Scheme

1: **Initialize** $[T^{v,n}=0]_{V\times N}$, replay memory buffer $[\mathcal{D}^{v,n}]_{V\times N}$, γ
2: **for** $t=1$ to t_{max} **do**
3: **for all** $\text{VS}_{v,n}$ that needs an association decision **do**
4: Get $s_{v,n}^{T^{v,n}}$ and calculate $R_{v,n}^{T^{v,n}-1}$.
5: Store ($s_{v,n}^{T^{v,n}-1}$, $a_{v,n}^{T^{v,n}-1}$, $R_{v,n}^{T^{v,n}-1}$, $s_{v,n}^{T^{v,n}}$) in $\mathcal{D}$.
6: Select a random action $a_{v,n}^{T^{v,n}}$ with probability ϵ.
7: Otherwise $a_{v,n}^{T^{v,n}} = \arg\max_{a_{v,n}^{T^{v,n}}} Q(s_{v,n}^{T^{v,n}}, a_{v,n}^{T^{v,n}}; \boldsymbol{\theta}_{v,n})$.
8: Sample a mini-batch from $\mathcal{D}$.
9: Update $\boldsymbol{\theta}_{v,n}$ by minimizing (4) via a gradient descent algorithm.
10: $T^{v,n} = T^{v,n}+1$, and replace $\hat{\boldsymbol{\theta}}_{v,n}$ with $\boldsymbol{\theta}_{v,n}$ periodically.
11: **for all** (m,n) that need to change the SC allocation scheme at t **do**
12: **Initialize** $\mathcal{F}_0^{m,n}$, $\psi_0^{m,n}$, $\lambda=0$.
13: **for** $\lambda=1$ to λ_{max} **do**
14: **if** $\lambda\%2=0$ **then**
15: Randomly select a player $L_k^{m,n}$ and $L_k^{m,n}\in F_i^{m,n}$.
16: Randomly select a coalition $F_j^{m,n}$ and $j\neq i$.
17: **if** $F_j^{m,n}\succ_k F_i^{m,n}$ **then**
18: $F_i^{m,n}=F_i^{m,n}\backslash L_k^{m,n}$, $F_j^{m,n}=F_j^{m,n}\cup L_k^{m,n}$.
19: **else**
20: Randomly select a player $L_{k_1}^{m,n}$.
21: **for** $L_{k_2}=1$ to C **do**
22: **if** L_{k_1} and L_{k_2} is a swap-blocking pair **then**
23: $\psi^{m,n}=\psi_{k_2 j}^{k_1 i}$.
24: **if** $t-t_{update}=\delta$ **then**
25: All VSs upload local models. $t_{update}=t$.
26: **for** $n=1$ to N **do**
27: Update $\boldsymbol{\theta}_n^*$ by (5), and broadcast it to all VSs.
28: $t=t+1$.

where γ is the discount factor. The training process of DDQN is detailed in lines 3-10 of **Algorithm 1**. It should be noted that since the agent's reward involves the average performance over multiple short timeslots, the reward and experience of the $(T-1)$-th long timeslot can only be obtained at the beginning of the T-th long timeslot (lines 4-5).

In the proposed FDRL framework, each VS needs to periodically sends its local model parameters to the BS to update the global model. When the BS receives the local models uploaded by all VSs, it calculates the global model for each service type [3]:

$$\boldsymbol{\theta}_n^* = \frac{\sum_{v=1}^V a_{v,n} E_{v,n} \boldsymbol{\theta}_{v,n}}{\sum_{v=1}^V a_{v,n}}, \quad (5)$$

where $a_{v,n}$ indicates whether service n in vehicle v is activated or not. $E_{v,n}$ is the amount of training data of $\text{VS}_{v,n}$. The process of FDRL is shown in lines 24-27 of **Algorithm 1**.

B. Alternating Coalitional Game and Swap-Matching Game Algorithm (ACGSM) for SC Allocation in Short-Timescale

Due to interference issues, the SC allocation schemes between BSs influence each other. To address this, we introduce an asynchronous decision approach, where only a subset of BSs update their SC allocation schemes in each short timeslot. As for which BSs can act simultaneously, we transform this problem into a graph coloring problem. Specifically, the BSs are treated as vertices of the graph, and if there is interference between two BSs, the corresponding vertices are considered adjacent. We use a backtracking method to obtain the optimal solution for this graph coloring problem.

$\mathcal{P}2$ is a 0-1 nonlinear integer problem, which is NP-hard. We propose an algorithm that alternates the coalitional game

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and swap-matching game to obtain an approximate optimal solution (lines 12-23 in **Algorithm 1**). In even-numbered iterations, the coalitional game is executed, while in odd-numbered iterations, the swap-matching game is applied. These two algorithms are performed alternately until convergence. Next, we will introduce these two algorithms in detail.

First, we model the SC allocation problem as a coalitional game. The player set comprises SCs allocated to the NS, i.e., $\mathcal{L} = \left\{L_k : k \in \mathbb{Z}, L_k \in \mathcal{C}, z_{m,n}^{L_k} = 1\right\}$. The coalition structure is defined as $\mathcal{F} = \{F_1, \ldots, F_{\sum_{v=1}^{V} y_{v,n}^m + 1}\}$, which contains all the VSs associated with the NS and one empty coalition. Each player can choose to join one coalition, meaning this SC is allocated to the corresponding VS. SCs that join the empty coalition are not allocated to any VS. All coalitions are disjoint, i.e., $F_i \cap F_j = \varnothing$ for any $i \neq j$, and $\cup_{i=1}^{|\mathcal{F}|} F_i = \mathcal{L}$. The characteristic function ($\mathcal{O}$) of this coalitional game is the objective function of $\mathcal{P}2$. Assuming that player L_k belongs to coalition F_i, if L_k leaving F_i to join coalition F_j can improve the utility $\mathcal{O}$, it is considered that player L_k prefers coalition F_j over coalition F_i. In this case, L_k needs to be transferred to F_j. The transfer rule is described as

$$F_j \succ_k F_i \Longleftrightarrow \mathcal{O}\left(\mathcal{F}'\right) > \mathcal{O}\left(\mathcal{F}\right), \tag{6}$$

where $\mathcal{F} = \{F_1, \ldots, F_i, \ldots, F_j, \ldots, F_{|\mathcal{F}|}\}$ and $\mathcal{F}' = \{F_i \backslash L_k, F_j \cup L_k, \mathcal{F} \backslash (F_i \cup F_j)\}$ [6]. When no player can obtain a higher system utility by unilaterally changing its coalition, the current coalition is considered to be a Nash-stable structure [6]. The process of the coalitional game is given in lines 14-18 of **Algorithm 1**. Upon the turn of coalitional game, we first need to randomly select player L_k and identify its coalition F_i (line 15). Then, another coalition F_j is randomly chosen, and the condition for player L_k to transfer to coalition F_j is checked (line 16). If it meets, then execute the coalition transfer action (lines 17-18).

Then, we model the SC allocation problem as a many-to-one matching problem between SCs and VSs, and introduce the swap-matching game to solve it. Specifically, the swap-matching is a matching generated by a swap operation, where two SCs swap their matched VSs while the matching schemes for other SCs remain unchanged [9]. The set of SC players is still denoted by $\mathcal{L}$, and the set of VS players is denoted by $\mathcal{B}$. The utilities of both SC players and VS players are defined as the characteristic function in the coalitional game, i.e., $\mathcal{O}$. A matching is represented by ψ, and the swap-matching is represented as $\psi_{k_2 j}^{k_1 i} = \psi \backslash \{(L_{k_1}, B_i), (L_{k_2}, B_j)\} \cup \{(L_{k_1}, B_j), (L_{k_2}, B_i)\}$, where (L_{k_1}, B_i) denotes that L_{k_1} matches with B_i. By introducing the concept of swap-blocking pair, we define the conditions under which the swapping operation is performed:

Definition 1: For a matching ψ, L_{k_1} and L_{k_2} is a swap-blocking pair if and only if: 1) The utilities of players involved in the swap operation are not reduced after the swap-blocking pair swap their matched players; 2) At least one of the players involved in the swap operation has an increase in utility.

A matching is two-sided swap-stable if and only if there are no swap-blocking pairs [9]. The steps of the swap-matching game are outlined in lines 19-23 of **Algorithm 1**. When it is time for swap-matching game, the first step is to randomly select an SC player L_{k_1} (line 20). Then, we iterate through all other SC players to check if there exists another player L_{k_2} that can form an swap-blocking pair with L_{k_1} (line 21). If such a player exists, L_{k_1} and L_{k_2} exchange their matched VS players (lines 22-23).

IV. Performance Evaluation

We design a circular simulation area with a diameter of 600 m, where vehicles move from the edge of the circle towards the center and continue until they reach the opposite edge. This design allows the DRL agent to experience a more diverse environment during training, improving its adaptability across different regions. Six BSs are randomly deployed within the simulation area, with each equipped with 16 SCs, each with a bandwidth of 0.1 MHz. The path loss model is set as $P_L^{m,n} = 36\log_{10} d_v^m + 20\log_{10} f_c + 7.6$ [10], where d_v^m is the distance between BS m and vehicle v, and $f_c = 2.6$ GHz is the carrier frequency [10]. Shadow fading follows a log-normal distribution with a standard deviation of 5 dB [1]. The transmission power of BS is 43 dBm [10], which is allocated to all SCs equally. The noise spectral density is set to -174 dBm/Hz [10]. We consider two NSs, with their QoS requirements specified as $(r_1^{min}, r_1^{max}, r_2^{min}, r_2^{max}) = (3, 15, 1, 5)$ Mbps, $(d_1^{min}, d_1^{max}, d_2^{min}, d_2^{max}) = (0.5, 1, 0.05, 0.2)$ ms, and $(w_1^r, w_2^r) = (0.8, 0.2)$. The packet sizes are $(b_1, b_2) = (500, 800)$ bits. Except for BS2, which has only NS1, and BS3, which has only NS2, all the other BSs have both two NSs, with each NS has 8 SCs. The vehicles' speeds follow a uniform distribution within $[5, 30]$ m/s, with a 0.01% chance of change in each short timeslot. The short-timescale is 0.01 s, and the maximum long-timescale is $\Delta t_{max} = 0.2$ s. The HO cost is set to 0.2. Besides, $(\alpha, \beta, w_f) = (10, 3, 0.2)$.

There are 6 vehicles involved in the simulation, with each vehicle having two types of services. To simulate dynamic service scenarios, we set $\text{VS}_{3,1}$ and $\text{VS}_{2,2}$ to be silent at the 2000th and 5000th long timeslots, respectively, and reactivate them after 5000 and 8000 long timeslots. DDQN is trained with a learning rate of 0.001, a batch size of 64, a discount factor of 0.99, and a replay memory of size 40000. The maximum iterations and model aggregation interval are set to 40000 and 200 long timeslots, respectively. To compare the proposed FDRL with conventional multi-agent DRL (MDRL), we record the change in accumulated reward values every 500 long timeslots, which are set as one epoch. Fig. 2 shows the performance of the first three vehicles. It can be seen that the proposed FDRL converges more quickly and achieves better convergence performance than MDRL. Furthermore, even though $\text{VS}_{3,1}$ and $\text{VS}_{2,2}$ are silent for some periods, resulting in insufficient training data in the early stages, they are able to receive the global model upon reactivation, thereby achieving rapid convergence. In contrast, their convergence in the MDRL algorithm is relatively slower.

Fig. 3(a) evaluates the impact of the number of candidate BSs ($|\mathcal{K}_{v,n}^T|$), using the average accumulated utilities of each service as the metric. We compare the proposed VSASA with the following schemes: 1) ANBS (associate to nearest BS): each VS connects to the nearest BS, without considering other factors; 2) WO_HC (without HO cost consideration): the same association algorithm as our approach but without

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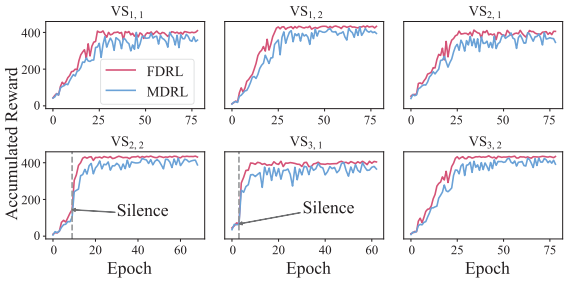


Fig. 2. Changes in the accumulated reward of each VS in the training process.

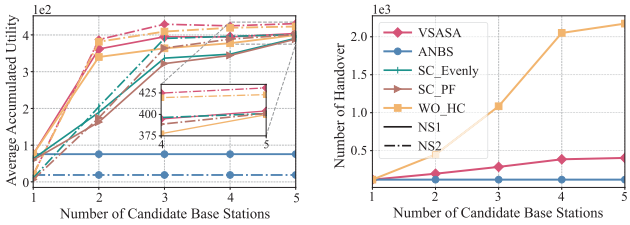


Fig. 3. Performance comparison in terms of (a) average accumulated utility and (b) number of handover with different number of candidate BSs.

considering HO cost in the reward function; 3) SC_Evenly (allocate SC evenly): each BS allocates its SCs equally among all VSs; 4) SC_PF (allocate SC by proportional fairness): SCs are allocated based on the proportional fairness algorithm, prioritizing VSs with lower current data rates. We observe that, apart from ANBS, the utility of other schemes increases with the number of candidate BSs. The reason is that more candidate BSs means that the VS has a greater likelihood of associating to a better BS. Since the list of candidate BSs is determined based on distance, once the number of candidate BSs reaches a certain level, the additional BSs provide limited gain. This explains why the increase in utility becomes less significant once the number of candidate BSs reaches 3. SC_Evenly and SC_PF differ from VSASA in their SC allocation schemes. The SC allocation scheme used in VSASA (ACGSM) yields higher utility, highlighting the importance of optimizing SC allocation. Since ANBS ignores factors such as load, HO cost, and available resources, it has the worst performance. In addition, we present the number of HOs for VSASA, ANBS, and WO_HC in Fig. 3(b). ANBS has the lowest number of HOs, which is reasonable because ANBS does not change the associated BS unless there is a significant change in the vehicle's location. Compared to WO_HC, VSASA can effectively reduce the number of HOs, which reflects the need to consider the HO cost.

In Fig. 4, we compare the proposed ACGSM with 1) CG (coalitional game): solely using coalitional game to determine SC allocation, and 2) OS (optimal solution): using exhaustive search to get the optimal solution. We fixed the locations of all vehicles and focused on the utility of NS1 in BS6, while keeping the SC allocation schemes of other BSs unchanged. We set the number of VSs to be 2 and vary the number of SCs to be 3 to 10 to obtain the results of the average utility of VSs shown in Fig. 4(a), while set the number of SCs to be 16 and vary the number of VSs to be 1 to 5 to obtain the results shown in Fig. 4(b). Compared to CG, ACGSM is closer to the optimal solution. Under different SC counts, the average deviations of ACGSM and CG from OS are 0.2% and

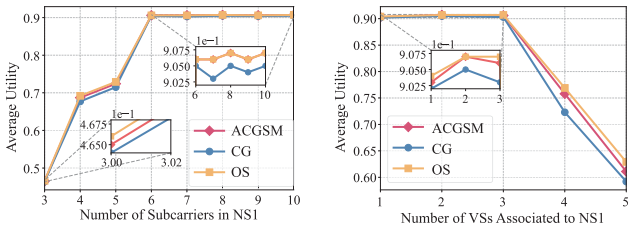


Fig. 4. Performance comparison in terms of the average utility of NS1 in BS6 with different (a) number of SCs and (b) number of VSs.

0.7%, respectively. Similarly, under varying VS counts, these deviations are 0.9% and 2.5%. Therefore, we can conclude that the proposed ACGSM can effectively improve algorithm performance and obtain near-optimal solutions.

V. Conclusion

In this letter, we propose a joint VS association and SC allocation scheme based on two-timescales in VSN-DMS. Specifically, in long-timescale, we design an FDRL-based VS association algorithm, which successfully improves the long-term utility of VSs while preserving vehicles' privacy. In short-timescale, we develop the ACGSM algorithm to address the SC allocation issue. ACGSM imposes no restrictions on the form of the objective function and can achieve near-optimal solutions. Finally, we validate the convergence advantages of FDRL, the superiority of the proposed VS association scheme, and the effectiveness of the SC allocation algorithm.

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